Phi Hung Nguyen

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Education

University of California, Los Angeles (UCLA) Expected Graduation: Dec. 2025
Bachelor of Science in Computer Science

Moorpark College Graduated May 2023
Associate of Science in Computer Science, Mathematics, and Physics | GPA: 4.0 - Dean's List

Technical Skills and Certificates

Languages: Python, C, C++, JavaScript, HTML
Development and Visualization Tools: Git, Shell Scripting, Visual Studio Code, Tableau
Libraries, Frameworks: React, Node.js, Firebase, Pandas, NumPy, Keras, TensorFlow, Scikit-learn, PyTorch, LoRA
Certificate: Machine Learning Foundations from Cornell University, Break Through Tech (View Certification)

Experience

Qualcomm Aug 2024 – Dec 2024
ML/AI Engineering Fellow *Los Angeles, CA*

- Developed and fine-tuned a Computer Vision model using PyTorch that achieved **85%** accuracy in weapon and crime detections as well as improving captioning accuracy by **30%**.
- Streamlined data preparation on a dataset of **1000+ images** by utilizing **data augmentation** and **tokenization** techniques to align multi-modal image-text data for vision-language tasks as well as rebalancing the classes for minimal bias in training.
- Employed **AdamW optimizer** and **Beam Search Algorithm** which led to robust convergence, **90%** precision in drawing bounding boxes, and improved quality of generated outputs for captions and responses in visual question answering (VQA) tasks by **25%**.
- Integrated **Low-Rank Adaptation (LoRA)** and **Transfer Learning** to boost model adaptability while retaining previously learned information, enabling accurate detections on new datasets without the need for additional training.

Break Through Tech AI UCLA May 2024 – Present
ML/AI Fellow *Los Angeles, CA*

- Selected from over 3000 applicants to participate in an intensive 12-month program mastering end-to-end data analysis pipelines, focusing on real-world applications across multiple domains such as **Image Recognition**, **Computer Vision**, and **Natural Language Processing**.
- Applied advanced Neural Network models (**Keras**, **Tensorflow**) to solve tasks like regression and visual/textual data processing, achieved predictive accuracy and computational efficiency of **85%** .
- Streamlined workflows for data preparation, modeling, and performance evaluation using industry-standard tools (**NumPy, Pandas, Scikit-learn**), achieving reproducible results across multiple domains.

Projects

CLASSYNC — Software Engineer | GitHub Repository — View Website Jan. 2024 – Mar. 2024

- Led a team of 5 to create an alternative UCLA Degree Audit System tailored for CS students.
- Architected and optimized the Firebase Real-time Database for efficient data retrieval and scalability, integrating it seamlessly with React components for a responsive user experience.
- Designed and implemented a secure, persistent authentication system using Firebase Authentication, ensuring data integrity and owner-specific accessibility.
- Deployed the finished build online, ensuring smooth functionality and accessibility across platforms.

IRON MAN Gauntlet - Hack Competition — Electrical Engineer | GitHub Repository Jul. 2023

- Led a team of 3 to design a gauntlet within 48 hours, enabling real-time hazard detection and data transmission.
- Developed an **Arduino-based circuit** with ultrasonic, air quality, and temperature/humidity sensors, **achieving 92% detection accuracy** and real-time hazard alerts via an OLED display and adaptive LED ring, which earned the **"Most Innovative Electrical Design" award** for the LED ring's hazard-responsive color display.
- Collaborated on CAD and web development to enable seamless data transfer from the gauntlet to the website, culminating in both successful run in the test field and design presentation to a panel of judges.